

Riphah International University Islamabad

I-14 Campus

Department of Artificial Intelligence

PROJECT PROPOSAL

Student Management System

Submitted By:

- Muhammad Salman (SAP ID: 72373)
- Shaheryar Khan (SAP ID: 72775)

Submitted To:

Prof. Muhammad Imran

Course:

Object Oriented Programming (OOP)

Semester:

2nd Semester

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1. Introduction

Student Management System is a Java-based Object Oriented Programming project developed to manage student and teacher information in an organized way. This project demonstrates the practical implementation of OOP concepts such as Inheritance, Polymorphism, Abstraction, and Encapsulation.

The system allows users to display student information, teacher information, course details, attendance, result, fee information, and notifications through a menu-driven interface. The project also provides dynamic user input functionality where users can manually enter student, teacher, and course details.

This project is designed for educational purposes and helps students understand how classes, objects, constructors, inheritance, and method overriding work together in real-world applications.

2. Objectives of the Project

The main objectives of this project are:

- To understand the practical implementation of Object Oriented Programming.
 - To develop a menu-driven Java application.
 - To manage student and teacher records efficiently.
 - To implement all four pillars of OOP.
 - To improve programming and problem-solving skills.
 - To create a simple and user-friendly management system.
-

3. Features of the Project

The Student Management System provides the following features:

- Display Student Information
- Display Teacher Information
- Display Course Information
- Display Student Result and Grade
- Show Attendance Percentage
- Display Department Information
- Fee Payment System
- Notification System

- Manual Data Entry System
- Menu Driven Interface

The project contains default stored data as well as manual input functionality for dynamic updates.

4. OOP Concepts Used in the Project

4.1 Inheritance

Inheritance is used when Student and Teacher classes inherit properties from the Person class.

Example:

```
class Student extends Person
class Teacher extends Person
```

This reduces code duplication and improves code reusability.

4.2 Abstraction

Abstraction is implemented using the abstract class Person and abstract method displayInformation().

```
abstract class Person
```

This helps in hiding implementation details and enforcing structure in child classes.

4.3 Polymorphism

Polymorphism is achieved through method overriding.

Both Student and Teacher classes override the displayInformation() method according to their requirements.

```
@Override
void displayInformation()
```

4.4 Encapsulation

Encapsulation is implemented by declaring variables as private and accessing them through getter methods.

```
private int id;  
private String name;
```

This provides data security and controlled access.

5. Classes Used in the Project

The project contains the following classes:

1. Person (Abstract Class)
2. Student
3. Teacher
4. Course
5. Result
6. Attendance
7. Department
8. Fee
9. Notification
10. Main Class

Each class performs a specific task and contributes to the overall functionality of the system.

6. Methodology

The project is developed using Java programming language in an Object Oriented approach.

The development process includes:

1. Designing classes and relationships.
2. Implementing OOP concepts.
3. Creating constructors and methods.
4. Developing menu-driven functionality.
5. Testing user input and outputs.

6. Improving interaction through manual data entry.

The project uses Scanner class for user input and switch statements for menu navigation.

7. Expected Output

The system will provide an interactive menu where users can:

- View student details
- View teacher details
- View course information
- Check grades and attendance
- Pay fee
- Send notifications
- Enter student and teacher details manually

The project will display organized and readable outputs on the console.

8. Conclusion

The Student Management System is a simple but effective OOP-based Java project that demonstrates the practical use of Object Oriented Programming concepts. The project improves understanding of inheritance, abstraction, polymorphism, encapsulation, constructors, objects, and menu-driven systems.

This project also enhances programming skills and provides a foundation for developing larger management systems in the future.

9. Tools and Technologies

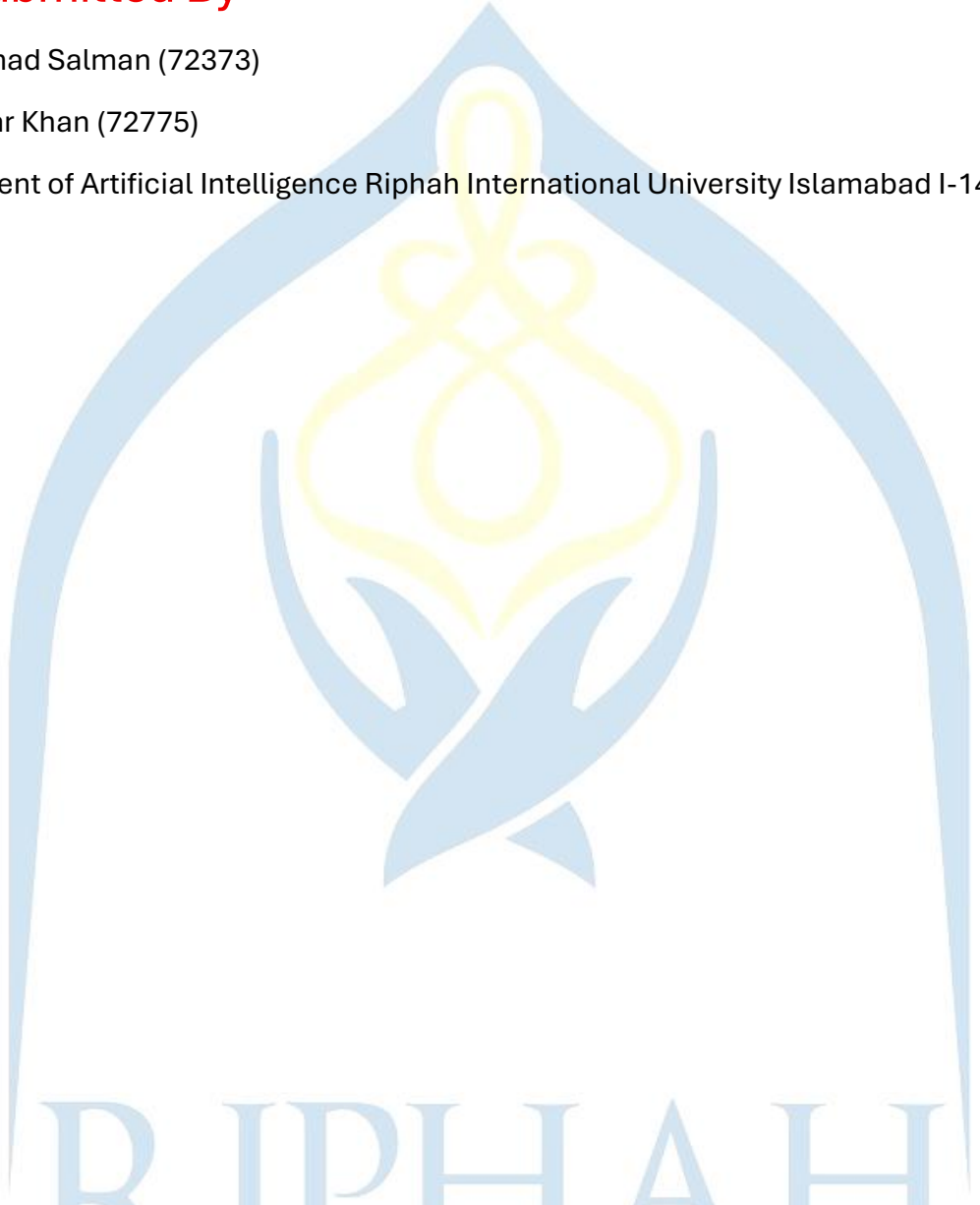
- Programming Language: Java
 - IDE: IntelliJ IDEA / NetBeans / Eclipse
 - Concepts Used: OOP, Classes, Objects, Constructors, Inheritance, Abstraction, Encapsulation, Polymorphism
-

10. Submitted By

Muhammad Salman (72373)

Shaheryar Khan (72775)

Department of Artificial Intelligence Riphah International University Islamabad I-14
Campus



The logo of Riphah International University is a large, light blue emblem. It features a stylized yellow flower or star-like shape in the center, composed of several interlocking loops. This central element is framed by a large, light blue arch that resembles a stylized 'R' or a protective canopy. Below the arch, the words 'RIPHAH INTERNATIONAL UNIVERSITY' are written in a large, light blue, serif font, stacked in three lines.

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